

ELECTRONIC DEVICES AND APPLICATIONS

I B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC03	ESC	L	T	P	C	CIE	SEE	Total
		2	-	-	2	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To introduce components such as diodes, BJTs and FETs.
2. To know the applications of devices.
3. To know the switching characteristics of devices.

COURSE OUTCOMES

Upon completion of the Course, the students will be able to:

1. Acquire the knowledge of various electronic devices and their use on real life.
2. Know the applications of various devices.
3. Acquire the knowledge about the role of special purpose devices and their applications.

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	-	-	1	1	-	-	-	-	1
CO2	3	2	3	-	-	2	1	-	-	-	-	1
CO3	3	3	3	-	-	2	1	-	-	-	-	1

UNIT - I	DIODES	CLASSES: 10
Diodes: Diode – Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch-switching times.		
UNIT - II	DIODE APPLICATIONS	CLASSES: 12
Diode Applications: Rectifier - Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier, Rectifiers with Capacitive and Inductive Filters, Clippers-Clipping at two independent levels, Types of Clampers.		
UNIT - III	BIPOLAR JUNCTION TRANSISTOR (BJT)	CLASSES: 09
Bipolar Junction Transistor (BJT): Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch, Need for Biasing, BJT as Amplifier.		
UNIT - IV	JUNCTION FIELD EFFECT TRANSISTOR (FET)	CLASSES: 10
Junction Field Effect Transistor (FET): Construction, Principle of Operation, Pinch-Off Voltage, Volt-Ampere Characteristic, Comparison of BJT and FET, FET as Voltage Variable Resistor, FET as amplifier.		
UNIT - V	SPECIAL PURPOSE DEVICES	CLASSES: 11
Special Purpose Devices: Zener Diode - Characteristics, Zener diode as Voltage Regulator, Operation of - SCR, UJT, Photo diode, Solar cell, LED.		

TEXT BOOKS

1. Jacob Millman - Electronic Devices and Circuits, McGraw Hill Education
2. Robert L. Boylestead, Louis Nashelsky Electronic Devices and Circuits theory, 11th Edition, 2009, Pearson.

REFERENCE BOOKS

1. Horowitz-Electronic Devices and Circuits, David A. Bell–5th Edition, Oxford.
2. Chinmoy Saha, Arindam Halder, Debaati Ganguly – Basic Electronics – Principles and Applications, Cambridge, 2018.